

Summary of Changes

We thank the reviewers for their constructive feedback. The revision implements all four items in our revision plan, together with several additional improvements prompted by the detailed reviews.

All line references below refer to the printed line numbers in the submitted diff file `revision-diff.pdf`, the revised manuscript with added text in blue and deleted material omitted.

The supplementary materials also contains an alternative diff file `revision-diff-full.pdf` showing deletions and has different line numbers, and an additional response to the reviewer C (`response-to-reviewer-C.pdf`).

1. Implementation of the Revision Plan

1. **Reframed the contribution.** The abstract, introduction, overview, and conclusion now frame the contribution as identifying and formalizing the Fixpoint Detection (FPD) gap between DBSP theory and implementation, and proving guarantees for the established internal-state-stability strategy—not proposing the strategy itself. See lines 9–19, 94–131, 265–311, and 1200–1216.
2. **Clarified the relationship with Feldera.** Sections 5, 6, and 9 relate IntConv and StFP to Feldera’s internal-state-stability strategy, distinguish that established idea from our formal guarantees, and explain our compositional treatment of nested bracketed circuits. We also qualified claims about the Feldera compiler version we inspected. See lines 659–669, 703–706, 780–796, 910–913, and 1162–1179.
3. **Narrowed the efficiency claims.** Section 5 now presents its example as a qualitative argument for a possible implementation benefit, not an empirical result or general lower bound. The conclusion identifies detector-overhead measurement and comparison with non-incremental recomputation as future work. See lines 732–772 and 1217–1222.
4. **Clarified the role of StFP.** Section 6 presents the StFP detector as a correctness-oriented fallback establishing decidability, and explains that implementations may instead use optimized or specialized detectors justified by Conv-completeness. See lines 922–930 and 933–938.

2. Additional Changes

- **Renamed the paper** to reflect the reframed contribution (lines 1–2).
- **Clarified the impossibility result.** We changed the terminology from “undecidability” to “impossibility of exact fixpoint detection” to clarify that the result rules out a computable runtime detector satisfying both soundness and completeness. Section 4.2 now states the circuit-level assumption that the chosen primitives make DBSP Turing-complete. It gives an explicit bounded-evaluator circuit and a concise proof using the non-recursive-enumerability of nonhalting, with a supporting citation. A footnote explains that the construction does not establish the same impossibility for incrementalizations of total computable queries. The section connects this limitation to IntConv and the later completeness guarantees for regular circuits and, separately, their incrementally optimized forms. The core nonhalting reduction is mechanized in Lean 4.

- **Refactored the overview.** Section 2 now distinguishes local FPD from global convergence detection, organizes the theory into specification, criterion, and detection layers, and introduces the principal concepts in their dependency order. See Fig. 4 and lines 263–311.
- **Expanded related work.** Section 9 now relates our finite-prefix FPD problem to prior work on derivatives of fixpoints and convergence of Datalog over pre-semirings, with corresponding bibliography updates. See lines 1154–1161 and references [2–3] at lines 1234–1239.
- **Improved precision and artifact correspondence.** The StFP definition now explicitly covers fixed continuations and its level-1, level-2, and vectorized cases (lines 797–832). A new paper-to-Lean map links the main definitions and results to the artifact.
- **Improved positioning and presentation.** Regular circuits are now described as a named fragment of the standard DBSP recursive sublanguage (lines 941–944 and 993–998). We also removed overclaims and duplicated text and corrected terminology, notation, and bibliography issues throughout; examples include the soundness qualifications at lines 707–731 and the Differential Dataflow discussion at lines 1180–1198.
- **Added statement of AI use.** See lines 1132–1135.